

Setup Guide: FreeRADIUS with daloRADIUS and Cisco Switch Configuration

October 9, 2024

1 Introduction

This guide provides detailed instructions for setting up a RADIUS server using FreeRADIUS and daloRADIUS, followed by the configuration of a Cisco switch to authenticate users via the RADIUS server. This setup allows centralized management of user authentication for network devices.

2 Wiring Description

The setup involves the following wiring configuration:

- A Linux-based server running FreeRADIUS and daloRADIUS is connected to the network. This server is connected to the Cisco switch via port 24, which is configured to operate on VLAN 99.
- A computer that will be used to test SSH access is connected to port 1 of the Cisco switch.
- The Cisco switch has port 24 assigned to VLAN 99 to communicate with the RADIUS server.

3 RADIUS Server Setup

3.1 Prerequisites

Ensure that you have a Linux-based server (e.g., Ubuntu) connected to the network. You will also need administrative privileges to install the required software.

3.2 Installing FreeRADIUS and daloRADIUS

To install FreeRADIUS and daloRADIUS, follow these steps:

1. Update the package list:

```
sudo apt update
sudo apt dist-upgrade
```

2. Install daloRADIUS using the provided installation script:

```
wget -qO - https://raw.githubusercontent.com/lirantal/
daloRADIUS/master/setup/install.sh | bash
```

3. Once the installation is complete, access daloRADIUS via a web browser using the server's IP address, e.g., `http://<server-ip>:8000`.

3.3 Configuring daloRADIUS

1. Log in to daloRADIUS using the default credentials (username: **Administrator**, password: <shown on daloRADIUS installation>).
2. Navigate to **Management** → **Users** and add a new user.
3. Set the **Service-Type** attribute for the user to **Administrative-User** to allow administrative access on network devices.
4. Navigate to **Management** → **NAS** and add the Cisco switch as a new NAS.
 - Set the IP address of the switch.
 - Enter the shared secret (must match the secret configured on the switch).
 - Enter a name for the switch

4 Cisco Switch Configuration

4.1 Configuring the Switch for RADIUS Authentication

Follow these steps to configure a Cisco switch for RADIUS authentication:

1. Connect to the Cisco switch via console or SSH.
2. Enter global configuration mode:

```
enable
terminal configuration
hostname MySwitch
ip domain-name example.com
crypto key generate rsa modulus 2048
ip ssh version 2
```

3. Configure AAA (Authentication, Authorization, and Accounting):

```
aaa new-model
aaa authentication login default group radius local
aaa authorization exec default group radius local
aaa accounting exec default start-stop group radius
aaa accounting network default start-stop group radius
```

4. Specify the RADIUS server and shared secret:

```
radius-server host 192.168.99.10 auth-port 1812 key MyRadiusSecret
```

Replace 192.168.99.10 with the IP address of your RADIUS server and MyRadiusSecret with the shared secret.

5. Configure VLAN and interfaces:

```
interface vlan 99
ip address 192.168.99.1 255.255.255.0
exit
```

```
interface FastEthernet0/24
switchport mode access
switchport access vlan 99
no shutdown
exit
```

```
interface FastEthernet0/1
switchport mode access
```

```
switchport access vlan 99
no shutdown
exit
```

6. Configure SSH access for users authenticated via RADIUS:

```
line vty 0 4
login authentication default
transport input ssh
exit
```

5 Testing the Configuration

1. Use an SSH client to connect to the switch from a computer connected to the network.
2. Enter the username and password that was configured in daloRADIUS.
3. Verify that the login is successful and that you have administrative access.
4. Create a new user in radius and verify that user can login, too

Switches have old ssh version, so you may need to run:

```
ssh -oCiphers=+aes256-cbc , aes192-cbc , aes128-cbc , 3des-cbc
-oHostKeyAlgorithms=+ssh-rsa -oKexAlgorithms=+diffie-hellman-group14-sha1
admin@192.168.99.1
```